$V = K_1 v_1 + K_2 v_2 + K_3 v_3$

Linear Dependence and Independence

A set of vectors is said to be linearly dependent if at least one of the vectors in the set can be defined as a linear combination of the others; if no vector in the set can be written in this way, then the vectors are said to be linearly independent.

Definition

If $S = \{v_1, v_2, v_3, \dots, v_n\}$ is a non-empty set of vectors in a vector space V, then the vector equation

$$k_1 v_1 + k_2 v_2 + k_3 v_3 + \cdots + k_n v_n = 0$$

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has at least one solution, namely,

$$k_1 = 0, k_2 = 0, \dots, k_n = 0.$$

We call this the trivial solution. If this is the only solution, then S is said to be linearly independent set. If there are solutions in addition to the trivial solution, then S is called as linearly dependent set.

Example 1: **Determine whether the set consists of vectors $v_1 = (1, -2, 3), v_2 = (5, 6, -1), v_3 = (3, 2, 1)$ are linearly dependent or independent.**

Solution: **Let $k_1 v_1 + k_2 v_2 + k_3 v_3 = 0$ be the linear combination. Then,**

$$K_1 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + K_2 \begin{pmatrix} 5 \\ 6 \\ -1 \end{pmatrix} + K_3 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 0$$

$$k_1(1, -2, 3) + k_2(5, 6, -1) + k_3(3, 2, 1) = 0$$

$$(k_1 + 5k_2 + 3k_3, -2k_1 + 6k_2 + 2k_3, 3k_1 - k_2 + k_3) = (0, 0, 0)$$

$$k_1 + 5k_2 + 3k_3 = 0$$

$$-2k_1 + 6k_2 + 2k_3 = 0$$

$$3k_1 - k_2 + k_3 = 0$$

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ -2 & 6 & 2 & 0 \\ 3 & -1 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 16 & 8 & 0 \\ 0 & -16 & -8 & 0 \end{array}\right) R_2 + 2R_1, R_3 - 3R_1$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 2 & 1 & 0 \end{array}\right) \frac{R_2}{8}, \frac{R_3}{-8}$$

$K_1 + 5K_2 - 6K_3 = 0$

$$\begin{pmatrix} 0 & 2 & 1 & | & 0 \end{pmatrix} \quad 8 \quad -8$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) R_3 - R_2$$

$K_1 + 5K_2 - 6K_2 = 0$

$K_1 - K_2 = 0$

$\boxed{K_3 = -2K_2}$

$K_1 = K_2$

$K_2 = t$

$$\begin{cases} k_1 + 5k_2 + 3k_3 = 0 \quad \dots\dots(1) \\ 2k_2 + k_3 = 0 \quad \dots\dots\dots(2) \\ 0 = 0 \end{cases}$$

$K_1 = t, \; K_2 = t, \; K_3 = -2t$

K

From equation (2), we get

$k_3 = -2k_2$

Putting value of k_3 in equation (1), we get:

$k_1 + 5k_2 + 3(-2k_2) = 0$

$k_1 - 1k_2 = 0$

$k_1 = 1k_2$

Let $k_2 = t$, therefore

$k_1 = 1t$

$k_3 = -2t$

As,

$k_1 = k_2 = k_3 \neq 0$

So, the vectors v_1, v_2, v_3 are linearly dependent.

Linearly dependent

Only ZERo LInear

Example 2: Determine whether the set consists of

vectors $v_1 = (1,0,1,2), v_2 = (0,1,1,2), v_3 = (1,1,1,3)$

are linearly dependent or independent.

$$K_1\begin{pmatrix}1\\0\\1\\2\end{pmatrix} + K_2\begin{pmatrix}0\\1\\1\\2\end{pmatrix} + K_3\begin{pmatrix}1\\1\\1\\3\end{pmatrix} = 0$$

$K_1 + 0K_2 + 1K_3 = 0$

$0K_1 + K_2 + K_3$

$K_1 + K_2 + K_3 = 0$

$2K_1 + 2K_2 + 3K_3 = 0$

$\Longrightarrow$

$K_1 + K_3 = 0$

$K_2 + K_3 = 0$

$K_1 + K_2 + K_3 = 0$

$2K_1 + 2K_2 + 2K_3 = 0$

$\cancel{K_1} + \cancel{K_3} = 0$

As
$K_1 = K_2 = K_3 = 0$ Only
The linearly
Independent

$$\begin{array}{r} \cancel{K_1} + \cancel{K_3} = 0 \\ -K_1 + K_2 + \cancel{K_3} = 0 \\ \hline -K_1 = 0 \end{array} \Rightarrow \boxed{K_1 = 0}$$

$0 + 2K_2 + 0 = 0$ $\boxed{K_2 = 0}$ $0 + K_3 = 0$ $\boxed{K_3 = 0}$

(1) **Exercise:** Determine whether vectors $v_1 = (1,0,0), v_2 = (0,1,0)$, $v_3 = (0,0,1)$ are linearly dependent or independent in R^3. 1, 3, 5, 7

(2) **Exercise:** Does $S = \{(1,2,3), (0,1,2), (3,0,-1)\}$ form a linearly independent set of vectors in R^3? 2, 4, 6, 8

$K_1 + 3K_3 = 0$
$2K_1 + K_2 = 0$
$3K_1 + 2K_2 - K_3 = 0$

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & -1 & 0 \end{array}\right)$$

$R_2 - 2R_1$
$R_3 - 3R_1$

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & -6 & 0 \\ 0 & 2 & -10 & 0 \end{array}\right)$$

$R_3 - 2R_2$

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & -6 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right)$$

K_1 K_2 K_3

$$R_3 - 4R_2 \begin{pmatrix} 0 & 0 & 2 & | & 0 \end{pmatrix}$$

$$\frac{1}{2} R_3 \quad \begin{array}{cccc} k_1 & k_2 & k_3 & \\ \end{array}$$
$$\begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & -6 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$k_3 = 0$

$k_2 - 6k_3 =$ $\quad k_2 = 0$

$k_1 + 3k_3 = 0 \quad k_1 = 0$

$$\begin{pmatrix} 1 & 2 & 3 & | & 0 \\ 1 & 1 & 2 & | & 0 \\ 2 & 0 & 2 & | & 0 \end{pmatrix}$$

$$\begin{array}{l} \\ R_2 - R_1 \\ R_3 - 2R_1 \end{array} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -4 & -4 & 0 \end{pmatrix}$$

$$\begin{array}{l} -R_2 \\ -\frac{1}{4}R_3 \end{array} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$R_3 - R_2 \quad \begin{array}{cccc} k_1 & k_2 & k_3 & \\ \end{array}$$
$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$k_2 + k_3 = 0 \quad k_2 = -k_3$

$k_1 + 2k_2 + 3k_3 = 0 \quad k_2 = -t$

$k_1 + 2t + 3t = 0$

$k_1 = \quad t$

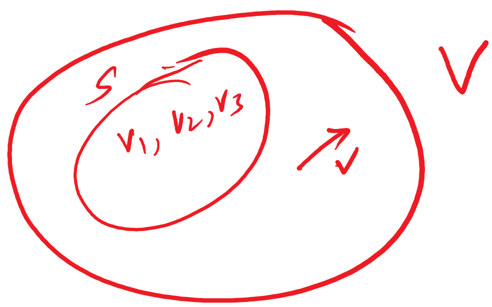
$k + t = 0$

Linearly dependent $(t, -t, t)$

Let S be the set of some vectors of vector space V. If every

vector in V is a linear combination of vectors in S, then the

set S is said to span V or the V is spanned by the set S.



DEFINITION 3

The subspace of a vector space V that is formed from all possible linear combinations of the vectors in a nonempty set S is called the ***span of S***, and we say that the vectors in ***S span*** that subspace.

If $S = \{w_1, w_2, w_3, \ldots, w_r\}$, then we denote the span of S by span $\{w_1, w_2, w_3, \ldots, w_r\}$ or $span(S)$.

Example1: Let $S = \{(1,0,0), (0,1,0), (0,0,1)\}$ be the set of standard unit vectors in R^3. Prove that S spans R^3.

Solution:

S spans R^3 means every vector of R^3 can be written as linear combination of these three vectors i.e.

$$(a,b,c) = k_1\overrightarrow{v_1} + k_2\overrightarrow{v_2} + k_3\overrightarrow{v_3}$$

$$(a,b,c) = k_1(1,0,0) + k_2(0,1,0) + k_3(0,0,1)$$

$$(a,b,c) = (k_1, 0,0) + (0, k_2, 0) + (0,0,k_3)$$

$$(a,b,c) = (k_1, k_2, k_3)$$

$a=3, b=-2, 4$

$(3,0,0) + (0,-2,0) + (0,0,4)$

$(3,-2,4)$

$a = k_1$, $b = k_2$, $c = k_3$

$=> (a,b,c) = a(1,0,0) + b(0,1,0) + c(0,0,1)$

e.g. $(1,2,3) = 1(1,0,0) + 2(0,1,0) + 3(0,0,1)$

Note: The Standard Unit Vectors Span R^n. How?

Recall that the standard unit vectors in R^n are
$e_1 = (1,0,0,\ldots,0), e_2 = (0,1,0,\ldots,0), \ldots, e_n = (0,0,0,\ldots,1)$.

These vectors span R^n since every vector $v = (v_1, v_2, \ldots, v_n)$ in R^n can be expressed as
$v = v_1e_1 + v_2e_2 + \ldots + v_ne_n$
which is a linear combination of $e_1, e_2, \ldots, e_n$. Thus, as in example 1, the vectors

$\boldsymbol{i} = (1,0,0), \boldsymbol{j} = (0,1,0), \boldsymbol{k} = (0,0,1)$

span R^3 since every vector in this space can be expressed as linear combination of these three vectors (also called basis).

Example2: Let V be the vector space R^3 and $\overrightarrow{v_1} = (1,2,1), \overrightarrow{v_2} = (1,0,2), \overrightarrow{v_3} = (1,1,0)$.
Check $\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}$ span R^3.

$$\begin{pmatrix} a \\ b \\ \end{pmatrix} = k_1\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + k_2\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + k_3\begin{pmatrix} 1 \\ 1 \\ \end{pmatrix}$$

Check v_1, v_2, v_3 span R^3:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + k_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{aligned} a &= k_1 + k_2 + k_3 \\ b &= 2k_1 + 0k_2 + 1k_3 \\ c &= k_1 + 2k_2 + 0k_3 \end{aligned}$$

$k_1 + k_2 + k_3 = a$ ...(1)

$2k_1 + 0 + k_3 = b$ ...(2)

$k_1 + 2k_2 + 0$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix} = 1(0-2) - 1(4) + 1(4) = -2 - \cancel{4} + \cancel{4} = -2$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 2 & 0 & 1 & b \\ 1 & 2 & 0 & c \end{array}\right), \quad \begin{matrix} R_2 - 2R_1 \\ R_3 - R_1 \end{matrix} \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -2 & -1 & b-2a \\ 0 & 1 & -1 & c-a \end{array}\right)$$

$2c - 2a + b - 2a$

$$2R_3 + R_2 \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -2 & -1 & b-2a \\ 0 & 0 & -3 & b-4a+2c \end{array}\right)$$

$$k_3 = \frac{-4a+b+2c}{-3}$$

$$\overrightarrow{v_1} = (1,1,2), \overrightarrow{v_2} = (1,0,1), \overrightarrow{v_3} = (2,1,3)$$

$$\begin{vmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 1(0-1) - 1(3-2) + 2(1-0) = -1 - 1 + 2 = 0$$

$(1 \quad 1 \quad 2 \quad | \; a$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 1 & 0 & 1 & b \\ 2 & 1 & 3 & c \end{array}\right)$$

$$\begin{array}{l} \\ R_2 - R_1 \\ R_3 - 2R_1 \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & -1 & -1 & c-2a \end{array}\right)$$

$$\begin{array}{l} \\ -R_2 \\ -R_3 \end{array} \left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & 1 & a-b \\ 0 & 1 & 1 & 2a-c \end{array}\right)$$

$2a - c$
$-a + b$

$$R_3 - R_2 \left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & 1 & a-b \\ 0 & 0 & 0 & a+b-c \end{array}\right)$$

① Linear Independent / dependent

$K_1 V_1 + K_2 V_2 + K_3 V_3 = 0$ if K_1, K_2, K_3 3×3

if we get Only ZERO LI (Linear Independent)

$K_1 = K_2 = K_3 \cdots K_n = 0$ Only

Then Linearly Independent

other case dependent

② $S(V_1, V_2, V_3)$ spans any V

$k_1 v_1 + k_2 v_2 + k_3 v_3 = (a, b, c)$

$$k_1 \begin{pmatrix} v_{11} \\ v_{12} \\ v_{13} \end{pmatrix} + k_2 \begin{pmatrix} v_{21} \\ v_{22} \\ v_{23} \end{pmatrix} + k_3 \begin{pmatrix} v_{31} \\ v_{32} \\ v_{33} \end{pmatrix} \neq \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

If $\underline{3 \times 3}$ Then or $(n \times n)$

if $\begin{vmatrix} v_{11} & v_{21} & v_{31} \\ v_{12} & v_{22} & v_{32} \\ v_{13} & v_{23} & v_{33} \end{vmatrix} = 0$, Then Not Spanning

$\neq 0$, <u>Spanning</u>